

Question number	Answer	Additional guidance	Mark
8(a)	An explanation that includes four of the following points: • substitution mutation results in {base / nucleotide} being {replaced / swapped / changed with another} (1) • (substitution mutation) could affect one amino acid (1) • change {in one amino acid / in glycine} could affect {triple helix / collagen secondary structure / collagen tertiary structure} (1) • H bonds would not form / fewer H bonds would form (1) • {H bonds / triple helix} involved in strength (1) • a larger R group would loosen the triple helix (1)	ACCEPT stop codon could be inserted ACCEPT shorter collagen if linked to stop codon IGNORE 3D structure	(4)

Question number	Answer	Additional guidance	Mark
8(b)	An answer that includes the following points: - parent with OI : Oo and parent without OI : oo (1) - possible genotypes of offspring shown as Oo and oo (1) - corresponding phenotypes of offspring shown as Oo has OI and oo does not have OI (1) 1 : 1 / 50 : 50 (probability) (1)	NB ecf for {wrong parental genotypes / different letters used} for max 3 marks ecf mp 3 and 4 if X and Y given for max 2 marks ACCEPT any sets of upper- and lower-case letters IGNORE refs to carriers ACCEPT 1 in 2 / 50% / half / $\frac{1}{2}$ / 0.5 / 2 in 4	(4)

Question number	Answer	Mark
*8(c)	Non-spec methods shown in diagram: Ultrasound - would not harm mother or fetus – as there are no needles penetrating the tissues (A) - would only be useful if the {damage to bones / fractures} showed up (D) - can only be used relatively late into pregnancy – may be too late to have an abortion (D) NIPT - would not harm mother or fetus – as there are no needles penetrating the tissues (A) - can be used early on in pregnancy – abortion would still be an option / more ethical to abort a less-developed embryo (A) - of no use if female does not know she is pregnant that early – too late for abortion (D) Cordocentesis - can only be used relatively late into pregnancy – may be too late to have an abortion (D) - risk of miscarriage – parents may not want to lose embryo / unethical if nothing wrong with embryo (D) On-spec methods: Chorionic villus sampling - can be used fairly early on in pregnancy – abortion would still be an option / more ethical to abort a less-developed fetus (A) - has a fast turnaround time – reduces stress of waiting - risk of miscarriage – parents may not want to lose embryo / unethical if nothing wrong with embryo (D) Amniocentesis - safer in later pregnancies than CVS – less likely to have a miscarriage (A) - risk of miscarriage – parents may not want to lose embryo / unethical if nothing wrong with embryo (D) PGD - allows an unaffected embryo to be implanted - avoids the risks associated with CVS and amniocentesis of miscarriage / avoids the possibility of a pregnancy being terminated (A) - ethical issues surrounding the unused embryos – because they are living humans (D) - successful pregnancy is not guaranteed – causing stress / financial implications (D) Applicable to all DNA testing methods (except ultrasound for some points) - provides a DNA sample of the fetus - that can be tested for the mutations (A) - the specific mutation could be identified – allowing more informed decisions about keeping the child (A) - other disorders may be identified – which family members may not want to know about (D)	(6)

	• other disorders may be identified – which allows more informed decisions about keeping the child (A) • possibility of false positive / false negative results – unnecessary terminations / birth of child that had the condition (D)	
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Level 1

1 mark = a relevant comment

2 marks = simple discussion of **both** an advantage **and** disadvantage of screening methods {in general / limited to one group of methods}

Level 2

3 marks = simple discussion of **either** advantages for both groups of methods **or** disadvantages for both groups

4 marks = simple discussion of **both** advantages for both groups of methods **and** disadvantages that cover both groups

Level 3

5 marks = as for 4 marks **plus** one extended discussion of **either** advantages or disadvantages

6 marks = as for 4 marks **plus** extended discussion of **both** advantages **and** disadvantages

Level	Mark	Descriptor
	0	No awardable content
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts/concepts. Consequences are discussed, which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of biological facts/concepts. Consequences are discussed, which are supported throughout by sustained linkage to a range of scientific ideas, processes, techniques or procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.